

Synchronized Hand Difference Visualization for Piano Learning

Ruofan Liu
liu.r.ac@m.titech.ac.jp
Tokyo Institute of Technology
Japan

Erwin Wu
wu.e.aa@m.titech.ac.jp
Tokyo Institute of Technology
Japan

Chen-Chieh Liao
liao.c.aa@m.titech.ac.jp
Tokyo Institute of Technology
Japan

Hayato Nishioka
nishioka@csl.sony.co.jp
Sony Computer Science Laboratories
Japan

Shinichi Furuya
furuya@csl.sony.co.jp
Sony Computer Science Laboratories
Japan

Hideki Koike
koike@c.titech.ac.jp
Tokyo Institute of Technology
Japan

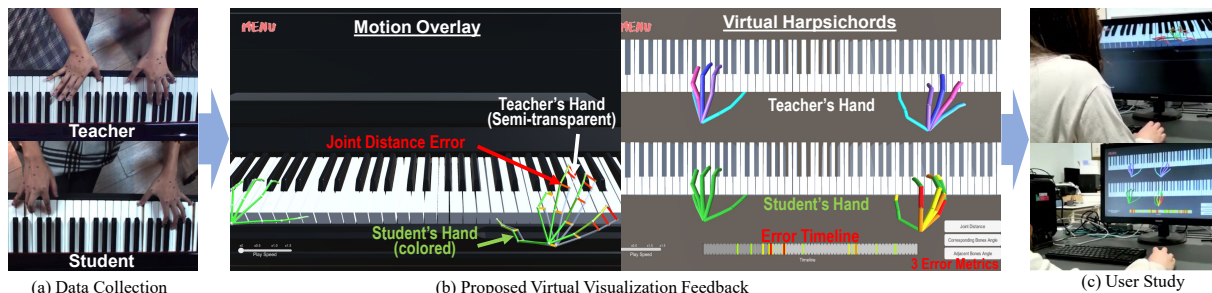


Figure 1: Overview of the piano learning system.

ABSTRACT

When learning a dexterous skill such as playing the piano, people commonly watch videos of a teacher. However, this conventional way has some downsides such as limited information to be retrieved and less intuitive instructions. We propose a virtual training system by visualizing differences between hands to provide intuitive feedback for skill acquisition. After synchronizing the data, two visual cues are proposed including a hand-overlay manner and a two-keyboards visualization. A pilot study confirm the superiority of the proposed methods over conventional video-viewing.

KEYWORDS

Piano Training, Motion Analysis, Virtual Visualization

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1 INTRODUCTION

Nowadays, a number of music educations rely on posture visualization such as videos or online lessons. However, conventional ways for piano learning have several inevitable problems. Most students

cannot play as fast as teachers so the videos are not temporally aligned. And previous methods lack auxiliary teaching functions, students are hence unable to quickly discover errors.

Therefore, we propose a new visualization system showing detailed differences in piano hand movements between students and teachers. After collecting the hand data through motion capture, we use the Temporal Cycle-Consistency (TCC) [Dwibedi et al. 2019] to temporarily synchronize the 3D data. We also implement two visualization approaches to display aligned data. The motion overlay method directly visualizes both teacher's and student's hands on a single keyboard. Another way is the virtual harpsichord method, resembling a harpsichord that separately exhibits two hands on two independent keyboards. To validate the proposed approaches, we conduct a pilot study. Several professional pianists are invited to experience our system. Most participants prefer the virtual harpsichord against the motion overlay and they can intuitively recognize when and where an error occur using this visualization.

2 RELATED WORK

Goebel et al. [Goebel and Palmer 2013] proposed a three-dimensional motion-capture system that could track the movements of finger joints and the hands of pianists. However, the data acquired were not aligned. TCC network [Dwibedi et al. 2019] could find correspondences between frames in multiple videos. Therefore, the per-frame embeddings retrieved from TCC network could be used to align 3D hand data. For visualization, previous studies suggested that motion analysis in a virtual environment could assist in skill transfer. Nozawa et al. [Nozawa et al. 2019] presented a VR ski training system, visualizing prerecorded teacher's motion in different ways for users to imitate. To help users specify particular moments, Kayukawa et al. [Kayukawa et al. 2018] illustrated the

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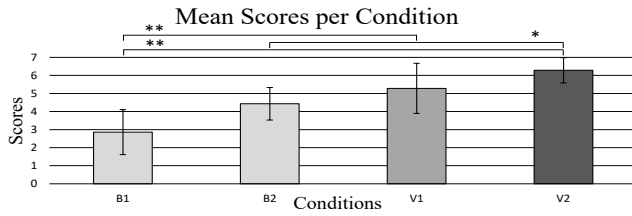


Figure 2: Quantitative results of the questionnaire. The brackets indicate the significant differences: * ($p < .05$), ** ($p < .01$).

elastic timeline. This was a useful way to show the frames when an error occurred, which we thereby implemented into our system.

3 IMPLEMENTATION

As a prototype, we collect data using marker-based motion capture. To handle unmatched playing speeds, we use TCC [Dwibedi et al. 2019] and a dynamic time warping algorithm to synchronize data. The main idea is to visualize the hand differences between two plays (student and teacher, they can also be different plays from the same person), we build two different approaches. Both methods have some common features such as a scroll-bar to adjust play speeds, and a controllable camera to observe hand movements from multiple perspectives.

3.1 Motion Overlay

The first visualization is the motion overlay, which is a straightforward approach to simply displaying both the teacher's and the aligned student's hands on the same piano keyboard.

In this condition (middle left in Fig.1), the student's and teacher's hand skeletons are in different colors for a better visualization. To provide intuitive feedback on the error between a student and a teacher, the distance between the corresponding joints is connected by a line. The colors of the distance lines, as well as each bone, are changed from green to red based on the magnitude of the error.

3.2 Virtual Harpsichord

However, some pianists suggest that instead of a "noisy" overlaid visualization, they demand a side-by-side option to compare each play individually. Thus, we implement a second visualization displaying the four hands on two separate keyboards, inspired by a harpsichord (piano with up and down keyboards, middle right in Fig.1). It allows the user to better see where the differences occur.

Besides the location of the error, we also hope the user can realize the timing of the error more intuitively. Therefore, an interactive timeline is added to visualize which segment of the entire clip the differences are happening. Based on error thresholds, the fault part will gradually turn from green to red. Also, to avoid too informative visualization, the errors are divided into three types: joint distance error, corresponding bone angle, and adjacent bone angle, which a user can choose from the bottom right checkbox (shown in Fig.1).

4 PILOT STUDY

The motion overlay is a straightforward and intuitive visualization, while the virtual harpsichord provides side-by-side comparison and clear error feedback. We run a pilot study to compare them with two baselines. The first baseline (B1) simply plays the two

original videos of the student and teacher. Since the videos are not synchronized, users cannot compare them simultaneously. Another baseline (B2) is the synchronized videos where the student's play speed is aligned with the teacher's. Our proposed methods are labeled as V1 (Motion Overlay) and V2 (Virtual Harpsichord).

Seven experienced pianists (6 female, 1 male, with experience ranging from 15 to 39 years.) are invited as participants. They are told to try each condition for 5 min and provide their overall preference for that condition in a 7-point Likert scale. Fig.2 shows the result of mean scores for each condition. An ANOVA statistical test suggests significant differences in the results so we conduct a Tukey's HSD post-hoc test. The result suggests that both the motion overlay and the virtual harpsichord are significantly better than the baseline of the two original videos. (V1-B1: $p=0.004$, V2-B1: $p=0.001$). When compared to the condition of two aligned videos, the virtual harpsichord are significantly better. (V2-B2: $p=0.033$).

5 DISCUSSION & CONCLUSION

From the pilot study, we can observe a trend that using temporally aligned videos is better. Also, both proposed methods outperform the two baselines which suggests the effectiveness of 3D visualizations. Between the two proposed methods, the majority of participants prefer the virtual harpsichord. From the later interview, we assume that this is because the side-by-side visualization and the error timeline provide better feedback on "where" and "when" the differences happen. However, this system is still a prototype with shortcomings. The current system relies on marker-based motion capture and cannot be used for online videos. Recent deep learning-based methods can perform 3D pose estimation from 2D videos which can be applied to our system in the future. Also, motion capture has some noises causing independent errors on the timeline. This may be overcome by adding a filter to preprocess the data.

Our system implements two visualization methods to help users compare hands of different plays. A pilot study with experienced pianists suggests that both proposed methods outperform conventional video-watching approaches. Between the two conditions, most people think the virtual harpsichord way can better help them with its clear feedback on errors. We believe this system can also be applied to other motor skills where a correct posture is essential.

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